Echo Yin

SENIOR A.I. ENGINEER

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I'm a senior AI engineer with over 10 years in software, the last several focused on machine learning and applied AI. I started out on enterprise systems, moved into online game programming and developer tooling, and since my master's in Computer Science I've concentrated on production AI: RAG pipelines, MCP tooling and agentic systems, most recently for a federal agency.

EXPERIENCE

Senior A.I. Developer · Impact Assessment Agency of Canada — Nov. 2025 – May 2026

- Collect and document clients' requirements and develop digital and infrastructure specifications for our A.I. enabled systems
- Design and coordinate the development, installation, integration and operation of our experimental A.I. systems
- Upgrade and develop Indigenous Consultation procedures for our Assessment Management System (AMS)

Online Programmer · Brass Lion Entertainment — Jul. 2023 – Apr. 2025

- Developed an item migration tool, optimizing item management by 96%
- Created a server log download tool, reducing QA bug reporting time by 95%
- Implemented a matchmaking system and data timer for server–side requests

Tools Engineer Intern · Rec Room — Jan. 2023 – Apr. 2023

- Built an in–house build download tool that reduced build download time by 70%
- Provided UI development and tooling support

MLOps Engineer Intern · PointClickCare — Jan. 2022 – Apr. 2022

- Supported MLOps data pipeline features and infrastructure
- Created comprehensive test cases for data pipeline validation
- Monitored and responded to security vulnerabilities

Software Engineer · Hubei Xinyegang Co., Ltd. — Jul. 2012 – Feb. 2020

- Handled software submission and maintenance cycles and led system analysis and review
- Assessed and executed process and system change requests, ensuring necessary data updates
- Supported external training programs through material creation and coordination

SELECTED PROJECTS

Feature ELM — github.com/e-choness/feature_extraction_cuda_elm

Extreme Learning Machines in C++20, trained in one least–squares solve on CPU or CUDA. C++ · CUDA · cuBLAS · cuSOLVER · CMake · GoogleTest · Docker · Gradio

Aegis AI Gateway — github.com/e-choness/aegis

A self–hosted AI gateway that shows its work. Python · FastAPI · LangGraph · Docker · PostgreSQL

EcoManage — github.com/e-choness/eco-manage

Energy management for real buildings. TypeScript · React · three.js · Node.js · MQTT · MongoDB · Redis · Docker

EDUCATION

AI Engineer Professional Certificate · Microsoft — Sep 2025 – Dec 2025

Master of Science – MS, Computer Science · Lakehead University · GPA 3.86/4.0 — Sep 2020 – May 2023

Bachelor of Engineering – BE, Software Engineering · Dalian Minzu University — Sep 2008 – Jul 2012

SKILLS

Machine Learning Frameworks PyTorch, TensorFlow/Keras, Scikit–learn, Hugging Face Transformers

Programming Languages Python, C++, C#, SQL **AI Specializations** NLP, Computer Vision, Large Language Models, Deep Learning

MLOps & Cloud Docker/Kubernetes, AWS SageMaker, Azure ML, MLflow

AI Infrastructure RAG Systems, MCP, Vector Databases, Model Fine–tuning